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Lent Term Project:

Factors Associated with Health Insurance Premiums

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SUMMARY

The study attempts to analyse whether some social, demographic, and geographical factors are associated with health insurance premiums. Through developing a Multiple Linear Regression Model and applying statistical theory and methods, we showed that **age, BMI, whether a health insurance contractor is a smoker, has children/dependents covered by the insurance, and lives in the North of the US or not** are among the factors that are likely to be associated with the amounts of the contracted insurance premiums in the US. It was further discovered that a synergy between an individual's age and smoking status was present. Having evaluated the limitations of our Multiple Linear Regression Model, we suggested possible improvement areas for further research.

BACKGROUND

Health insurance is considered to be essential for every household, as it is a safety net to fall back on when one becomes ill and brings an unexpected financial burden to a family. By buying health insurance policies, contractors agree to pay regular premiums, and they are guaranteed to receive the contracted amount of cash if certain unfortunate events occur within the contracted term.

Actuaries price the insurance products to help insurance companies stay afloat and make profits in the market by collecting premiums exceeding the total amount paid to the insured. To define the insurance premiums, the actuaries analyze a range of factors including data on contractors' socio-demographic characteristics, and normally, the premiums are higher for those contractors who are at a higher risk to fall sick in the future.

INTRODUCTION

The aim of the study is to understand how individuals' social, demographic, and geographical characteristics are associated with the levels of insurance premiums paid by them.

Cross-sectional data was sourced from Kaggle, a public data platform¹, from where we examined 1,338 observations to build and analyze several statistical models using the **Multiple Linear Regression Technique**. Using some variables, we constructed and analyzed four Candidate Models one by one. The **Final Model** was then selected amongst the **Candidate Models** and further improved to achieve the objective of our research.

For this study, we used data on individual medical costs billed by health insurance companies as our outcome variable **CHARGE**. We assumed that units for **CHARGE** are USD, for interpretational purposes, as Kaggle did not specify them. We further assumed that the medical costs billed by health insurance companies are correlated with the insurance premiums, the primary subject of our interest.

The Kaggle dataset also provided us with the information on six characteristics such as **AGE, BMI, CHILDREN, SEX, SMOKER** and **REGION**, which we chose to be the predictors, please refer to the **Table 1** below. Note that we assumed a primary beneficiary to be equivalent to a contractor of health insurance. We further assumed that dependents only include children.

¹ Choi, M. 2015, Medical Cost Personal Datasets, viewed 19 December 2021, <https://www.kaggle.com/mirichoi0218/insurance>

Table 1. Six Predictors Available for Multiple Linear Regression Analysis

#	PREDICTOR	DESCRIPTION
1	AGE	Age of primary beneficiary, years
2	BMI	Body mass index of primary beneficiary, kg/m2
3	CHILDREN	Number of children/dependents covered by health insurance: 0-5
4	SEX	Insurance contractor gender: female or male
5	SMOKER	Smoking status: yes or no
6	REGION	The beneficiary's residential area in the US: Northeast, Southeast, Southwest, Northwest.

In other words, in our study we attempted to answer several questions including but not limited to:

- Which social, demographic, and geographical characteristics are associated with higher/lower insurance premiums? Namely, are high premium payers, on average, older or younger? Do they tend to have high BMI or not? Do they tend to have more children? Do they smoke or not? Are more of them women or men? Do they live in certain geographic regions?
- How strong is the relationship between our response variable **CHARGE** and the six predictors such as **AGE**, **BMI**, **CHILDREN**, **SEX**, **SMOKER** and **REGION**? Note: the subset of statistically significant predictors is to be determined in the study.
- Is there synergy among the predictor variables?

ANALYSIS & DISCUSSION

To complete statistical analysis and reach the **Final Model**, which helps us answer the study objective, we undertook the following steps:

- EXPLORATORY DATA ANALYSIS (EDA):** EDA was carried out to investigate the dataset, relevant variables' plots as well as some descriptive statistics to guide the subsequent analysis.
- VARIABLE SELECTION:** Several variable selection methods were employed to determine the important predictor variables.
- ANALYSIS OF FOUR CANDIDATE MODELS:** Having independently created and assessed four **Candidate Models**, the **Final Model** that would best satisfy the objective of this study was agreed upon based on our judgement.
- FINAL MODEL DERIVATION & DIAGNOSTICS:** Further analysis using three statistical tools, namely, categorical variables' level reduction, introduction of interactions and applying transformations, were performed to finalize the **Final Model**. Outliers, leverage and influential points were examined to determine if the removal of any points would improve estimation accuracy. Next, adjusted R^2 was examined as part of Model assessment. Finally, residual analysis and multicollinearity checks were carried out to ensure that the underlying regression assumptions were met and the **Final Model** was sophisticated yet interpretable enough.
- FINAL MODEL INTERPRETATION:** The **Final Model**'s parameters were interpreted to verify that they make practical sense.
- LIMITATIONS OF THE FINAL MODEL & RECOMMENDATIONS:** Limitations of the **Final Model** were assessed with further improvement areas highlighted.

7. CONCLUSIONS were made.

1. EXPLORATORY DATA ANALYSIS

We started off by carrying out EDA to identify any issues with the initial Kaggle dataset, such as the presence of any obvious error points as well as any other data points that could potentially be removed in our further analysis.

First, **Figure 1** below shows the response variable **CHARGE** is highly positively skewed, where the median of \$9,382 is much smaller than the mean of \$13,270. Thus, in our study we will consider transforming **CHARGE**, possibly using the log transformation, and removing some data points (please refer to the Section 4.3 Transformations and Section 4.4 Outliers, Leverage and Influential Points). Moreover, from **Figure 1** we can see that outliers may be present in the data, potentially influencing the regression coefficients and impacting predictive accuracy.

Figure 1. Histogram of CHARGE

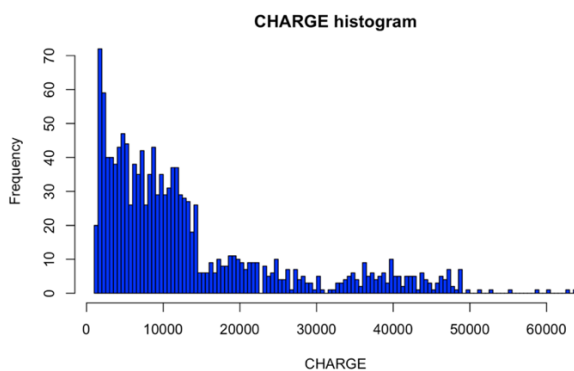
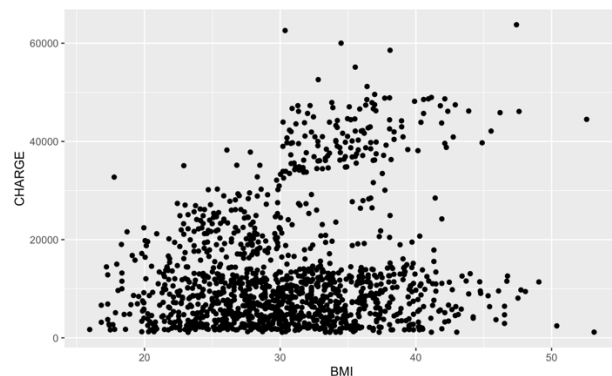


Figure 2. Scatterplot of CHARGE against BMI



Further, the individual scatter plots for **CHARGE** against the continuous independent variables, **BMI** and **AGE**, show that high leverage points may be present in the data.

In the case of the predictor variable **BMI**, we can see that there is a rough positive linear relationship present. However, the funnel shape of the data points suggested us to consider a transformation for the variable, please refer to **Figure 2**, for details see Section 4.3 Transformations.

For the scatter plot of **CHARGE** against **AGE**, there are data points with relatively large values in the direction of the x-axis, please see **Figure 3** below. We will deal with these high leverage points in the subsequent analysis (Section 4.4 Outliers, Leverage and Influential Points). Moreover, a closer examination of the plot shows us that the data lies in three roughly parallel upwards-sloping linear trends. This suggests that, in general, the older the contractors are, the higher the medical charges. However, this also indicates that there may be other variables that are responsible for the differences in medical charges for the same/similar age group. To determine such potential variable, we tried to plot **CHARGE** against **AGE** while accounting for **SMOKER**, please see **Figure 4**. The plot indeed suggests that there might be a factor interaction effect, which we discuss in Section 4.2 Interactions.

Figure 3. Scatter plot for CHARGE against AGE

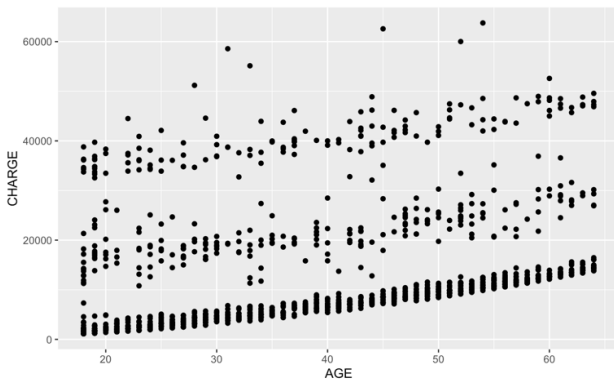
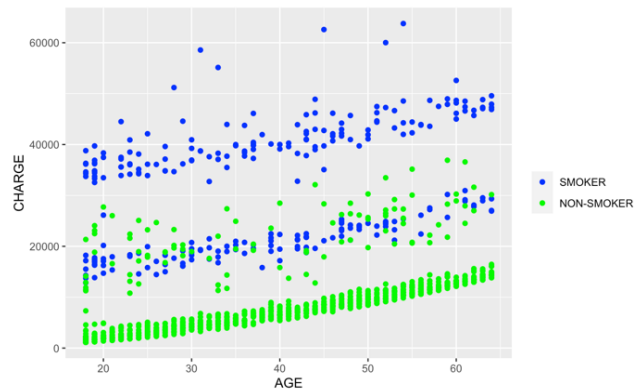


Figure 4. Scatter plot for CHARGE against AGE while accounting for SMOKER



2. VARIABLES SELECTION

In the **Initial Model** we included all six predictor variables which, later, we tested using four variables selection methods. **Table 2** provides more details on the six predictors.

Table 2. Six Predictors to be Used for Variables Selection

#	PREDICTOR	TYPE	(RE)CODING
1	AGE	Continuous	No need
2	BMI	Continuous	No need
3	CHILDREN	Categorical, 6 levels	Recoding : Five dummy variables created: CHILDREN_i = i if CHILDREN = i, 0 otherwise for i = 1, 2, 3, 4, 5.
4	SEX	Categorical, 2 levels	Recoding: 1 if SEX = female, 0 if male
5	SMOKER	Categorical, 2 levels	Recoding: 1 if SMOKER = yes, 0 if no
6	REGION	Categorical, 4 levels	Coding: three dummy variables created: REGION_NE = 1 if REGION = Northeast, 0 otherwise; REGION_SE = 1 if REGION = Southeast, 0 otherwise; REGION_SW = 1 if REGION = Southwest, 0 otherwise.

The following four selection methods were used to determine the statistically important predictors for the **Baseline Model**:

- forwards selection (based on AIC),
- backwards selection (based on AIC),

- c) stepwise selection (based on AIC) and
- d) best subsets regression (based on BIC and adjusted R²)

The first three methods - forwards, backwards and stepwise selection - suggested removal of one variable, **SEX**. The best subsets regression, under both criteria, also pointed that the predictor **SEX** should be excluded.

Hence, we excluded the variable **SEX**, and the **Baseline Model** based on the **five independent variables** was defined as follows:

$$CHARGE = \beta_0 + \beta_1 AGE + \beta_2 BMI + \beta_3 CHILDREN_1 + \dots + \beta_7 CHILDREN_5 + \beta_8 SMOKER + \beta_9 REGION_NE + \beta_{10} REGION_SE + \beta_{11} REGION_SW + \varepsilon$$

3. ANALYSIS OF FOUR CANDIDATE MODELS

Four **Candidate Models** were independently produced using a variety of statistical tools, e.g. transformations and interactions. The four models were then analyzed, please see **Tables 3 to 6** below, and the **Final Model** was agreed upon.

Table 3. Candidate Model 1 - REJECTED

BASELINE	CANDIDATE 1	MAJOR POINTS	OBSERVATIONS
CHARGE ~ AGE + BMI + CHILDREN + SMOKER + REGION	<u>log</u> (CHARGE) ~ <u>sqrt</u> (AGE) + <u>log</u> (BMI) + <u>KIDS</u> + SMOKER + <u>REGIONS</u>	CHILDREN and REGION were transformed into the binary variables KIDS and REGIONS , respectively. Please refer to Table 7 for more details.	Lower multiple R ² and adjusted R ² (0.765, 0.7641) compared to Candidate Model 4 (0.8018, 8009).

Table 4. Candidate Model 2 – REJECTED

BASELINE	CANDIDATE 2	MAJOR POINTS	OBSERVATIONS
CHARGE ~ AGE + BMI + CHILDREN + SMOKER + REGION	<u>log</u> (CHARGE) ~ <u>sqrt</u> (AGE) + <u>log</u> (BMI) + <u>KIDS</u> + SMOKER + <u>REGION NE</u> + <u>REGION SE</u> + <u>REGION SW</u>	Best subsets variables selection (using the adjusted R ² criterion) suggests removing the Northwest level for the categorical variable REGION . CHILDREN was transformed into the binary variable KIDS . Please refer to Table 7 for more details.	Removal of Northwest level of REGION results in incorrect interpretation of the model. Hence it cannot be done.

Table 5. Candidate Model 3 - REJECTED

BASELINE	CANDIDATE 3	MAJOR POINTS	OBSERVATIONS
CHARGE ~ AGE + BMI + CHILDREN + SMOKER + REGION	$\log(\text{CHARGE}) \sim$ $\sqrt{\text{AGE}} + \log(\text{BMI}) +$ $\text{CHILDREN} + \text{SMOKER}$ $+ \text{REGIONS}$	CHILDREN was assumed to be a continuous variable, as the values it can take are being relatively granular: integers from 1 to 5.	Assuming CHILDREN being a continuous variable is unrealistic.

Table 6. Candidate Model 4— ACCEPTED – FINAL MODEL

BASELINE	CANDIDATE 4	MAJOR POINTS	OBSERVATIONS
CHARGE ~ AGE + BMI + CHILDREN + SMOKER + REGION	$\log(\text{CHARGE}) \sim \text{AGE}$ $+ \log(\text{BMI}) + \text{KIDS} +$ $\text{SMOKER} +$ $\text{REGIONS} +$ $\text{AGE} * \text{SMOKER}$	CHILDREN and REGION were transformed into the binary variables KIDS and REGIONS , respectively. Please refer to Table 7 for more details. Interaction between AGE and SMOKER was found to positively improve the model. Please refer to Table 8 for more details.	Interpreting the binary variables KIDS and REGIONS may be less cumbersome than interpreting the original variables CHILDREN and REGION , respectively, with more than two levels.

The **Final Model** is as follows:

$$\log(\text{CHARGE}) = \beta_0 + \beta_1 \text{AGE} + \beta_2 \log(\text{BMI}) + \beta_3 \text{KIDS} + \beta_4 \text{SMOKER} + \beta_5 \text{REGIONS} + \beta_6 \text{AGE} * \text{SMOKER}$$

4. FINAL MODEL DERIVATION & DIAGNOSTICS

Here we analyze in extensive detail the steps taken to reach the **Final Model**, as well as the **Final Model** itself, to see if any further adjustments were needed.

4.1 CATEGORICAL VARIABLES' LEVEL REDUCTION

Let us examine the best subsets regression variables selection process in more detail. In addition to removing the predictor **SEX**, the best subsets regression recommends removing the Northwest level for the categorical variable **REGION**, under both criteria.

While accepting removing the predictor **SEX** from the **Initial Model**, we, however, could not freely remove a level of the categorical predictor **REGION** as it would result in wrong interpretation of the model. Thus, assuming the two Northern regions tend to have more in common and are together

different from the Southern regions in terms of health insurance costs, we decided to reduce the levels of the categorical variable **REGION** and transformed it into a binary predictor **REGIONS**. Another reason for doing this was that decreasing the levels of the variable may help increase interpretability of the statistical model.

Similarly, given that **CHILDREN** is a categorical variable with six levels, and it would be challenging to interpret it, we, re-grouped **CHILDREN** into a binary variable **KIDS** which represents whether an individual has any children/dependents covered by health insurance or not. Please see **Table 7** for further details.

Table 7. Reducing the Variables' Levels

ORIGINAL VARIABLE	RE-GROUPED VARIABLE	COMMENTS
CHILDREN , categorical, 6 levels	KIDS , categorical, 2 levels	Regroup CHILDREN into a binary variable KIDS : 1 if CHILDREN $\geq$ 1, 0 otherwise.
REGION , categorical, 4 levels	REGIONS , categorical, 2 levels	Regroup REGION into a binary variable REGIONS : 1 if REGION = northeast OR northwest, 0 otherwise.

Reducing levels of two categorical variables resulted in the following **Interim Model**:

$$CHARGE = \beta_0 + \beta_1 AGE + \beta_2 BMI + \beta_3 KIDS + \beta_4 SMOKER + \beta_5 REGIONS$$

4.2 INTERACTIONS

Three most promising interactions were analyzed and individually applied to the **Interim Model** based on the theoretical understanding of the variables, and the interaction **AGE*SMOKER** was finally chosen to be added to our **Interim Model**.

Tables 8 to 10 below demonstrate how we assessed the three promising interactions on the basis of the three criteria:

- increase in multiple R^2 ,
- increase in adjusted R^2 , and
- variance inflation factor (VIF) below 10, as introducing interaction terms may lead to multicollinearity.

Table 8. Interaction BMI*KIDS Analysis – REJECTED due to multicollinearity

BASELINE MODEL (No Interaction Term)	INTERACTION	MULTIPLE R^2	ADJUSTED R^2	VIF
Multiple R^2 : 0.7502, Adjusted R^2 : 0.7493, VIF: all values < 2	BMI*KIDS	0.7504	0.7493	KIDS: 26.3, BMI*KIDS: 28, all other VIFs < 3

Table 9. Interaction BMI*SMOKER– REJECTED due to multicollinearity

BASELINE MODEL (No Interaction Term)	INTERACTION	MULTIPLE R²	ADJUSTED R²	VIF
Multiple R ² : 0.7502, Adjusted R ² : 0.7493, VIF: all values < 2	BMI*SMOKER	0.8392	0.8385	SMOKER: 25.1, BMI*SMOKER: 25.5, all other VIFS < 2

Table 10. Interaction AGE*SMOKER – ACCEPTED

BASELINE MODEL (No Interaction Term)	INTERACTION	MULTIPLE R²	ADJUSTED R²	VIF
Multiple R ² : 0.7502, Adjusted R ² : 0.7493 VIF: all values < 2	AGE*SMOKER	0.7506	0.7495	AGE: 8.72, AGE*SMOKER:8.90, all other VIFS < 2

4.3 TRANSFORMATIONS

To ensure that the assumptions underlying the Multiple Linear Regression were met, and a model with a higher explanatory power was achieved, several transformations were applied to the response and continuous predictor variables.

We singled out those transformations, please refer to **Table 11**, which contributed positively to the five **Interim Model** characteristics:

- multiple R²,
- adjusted R², and
- linear relationship (measured by the absolute value of correlation) between the outcome variable **CHARGE** and the corresponding predictor.

Table 11. Introducing Transformations to Response and Continuous Variables

VARIABLE	Type	TRANSFORMATION	MAIN REASON FOR (NOT) TRANSFORMING
CHARGE	Response variable	log(CHARGE)	Applying log transformation to CHARGE will make its distribution less skewed (i.e., more symmetric, and bell-shaped). Please see the histogram of CHARGE in the R code in Appendix, Section 4.3 .
AGE	Continuous predictor	No transformation applied	- Correlation between log(CHARGE) and AGE is already relatively high (0.528) - No transformation applied as AGE is part of the AGE*SMOKER interaction term.

VARIABLE	Type	TRANSFORMATION	MAIN REASON FOR (NOT) TRANSFORMING
BMI	Continuous predictor	log(BMI)	The log transformation of BMI was selected among several BMI transformations (e.g., BMI ² , sqrt(BMI)) based on the absolute value of the correlation between log(CHARGE) and the transformed BMI .

Based on the transformations considered, the **Final Model** was reached:

$$\log(\text{CHARGE}) = \beta_0 + \beta_1 \text{AGE} + \beta_2 \log(\text{BMI}) + \beta_3 \text{KIDS} + \beta_4 \text{SMOKER} + \beta_5 \text{REGIONS} + \beta_6 \text{AGE} * \text{SMOKE}$$

4.4 OUTLIERS, LEVERAGE & INFLUENTIAL POINTS

Building up on the findings of the EDA, outliers, leverage, and influential points are examined in further detail. This is done to eliminate the possibility of conducting incorrect regression analysis, which will lead to inaccurate inferences being drawn.

Outliers & High Leverage Points:

Outliers were classified using standardised residuals (s.r.). Using the threshold s.r. value of 3, we detected 43 outlier observations. These were then individually examined in the Kaggle dataset to determine whether their removal had a significant impact on the robustness of the **Final Model**. The analysis revealed that they were not error points. As the outliers accounted for 3.2% of the data, their removal may result in loss of valuable information about the variability inherent in the dataset².

Therefore, to avoid distortion in inferences drawn, these outlier points were retained. The results of the analysis are in **Figure 5** below:

Figure 5. Plot for Standard Residuals

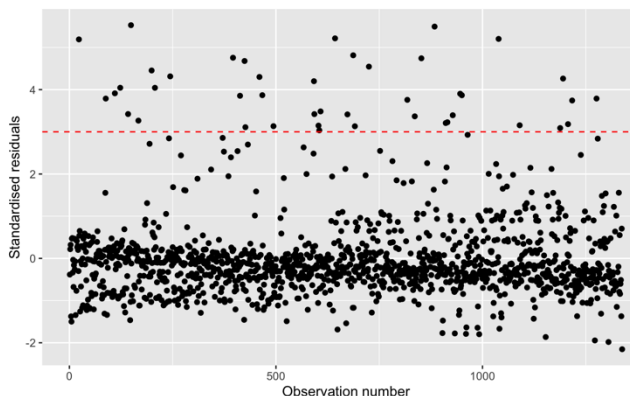
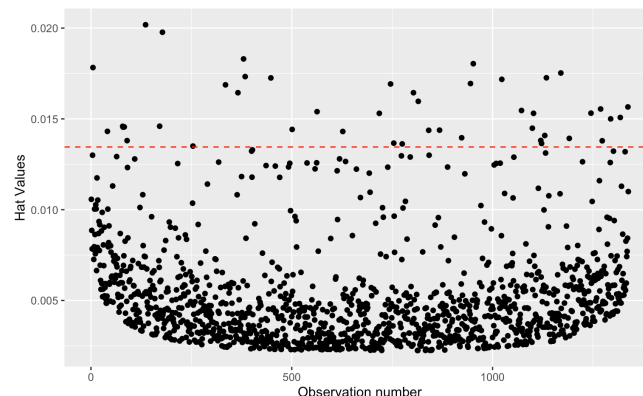


Figure 6. Plot for Hat Values



Next, the high leverage points were examined using the hat values. A threshold of $3 * \frac{p}{n}$ (≈ 0.013) was used to classify a data point as a high leverage point. Here, p represents the number of independent

² Frost, J. 2021, *Guidelines for Removing and Handling Outliers in Data*, viewed 29 December 2021, <https://statisticsbyjim.com/basics/remove-outliers/>

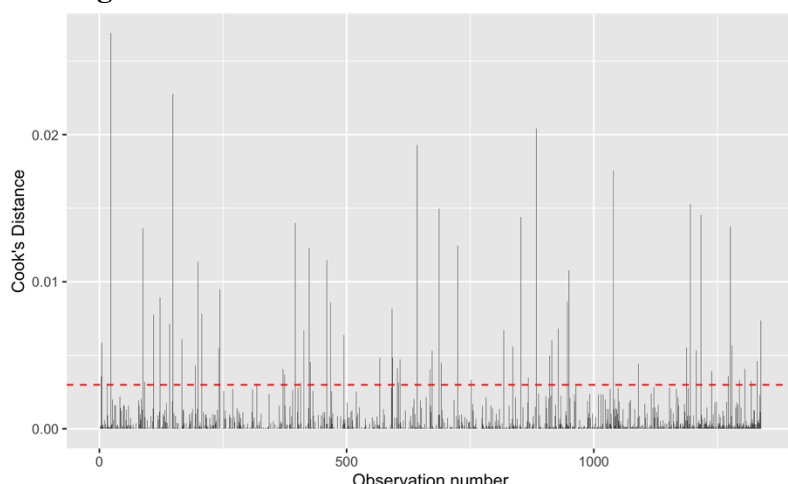
variables in the regression model and n represents sample size. We detected 45 high leverage points, as displayed in **Figure 6** above the red line. Similar to the outliers case, we chose to retain the high leverage points not to distort the inferences drawn from the **Final Model**.

Influential Points:

Finally, Cook's distance was calculated for each data point. A threshold of $\frac{4}{n}$ (≈ 0.003) where n is the sample size was then used to classify a point as an influential data point. We noted that more than 20 influential points are above the threshold value, please see **Figure 7** below. We chose to remove them as the influential points have the potential to significantly impact the regression model and distort its analysis.

Having refit the **Final Model**, we achieved a substantial increase in adjusted R^2 from 0.8009 to 0.9230.

Figure 7. Plot for Cook's Distance



4.5 ADJUSTED R^2 , RESIDUAL ANALYSIS & MULTICOLLINEARITY

Having implemented the analyses in Sections 4.1 - 4.4, the **Final Model** will now be analyzed further to determine whether the model obtained draws accurate inferences and to see if any further adjustments are required.

The adjusted R^2 value of 0.9230 indicates that 92.30% of the variation in **log(CHARGE)** can be accounted for by the proposed model, which is a promising figure.

Then, acknowledging that adjusted R^2 is not sufficient to evaluate the usefulness of the **Final Model** for practitioners, we proceed with performing residual analysis to examine the regression assumptions.

Table 12. Regression Assumptions Analysis

Ref.	REGRESSION ASSUMPTION	METHOD FOR TESTING	OUTCOME
A1	Linearity: The true model is linear in parameters	Plotting each independent variable against the residuals	Linearity is present as the plots for the outcome variable against the continuous independent variables are fairly random.

Ref.	REGRESSION ASSUMPTION	METHOD FOR TESTING	OUTCOME
A4	Homoscedasticity (the variance of error term, ϵ , is the same for all values of the predictors)	Plotting the residuals against the fitted values	The issue of heteroscedasticity is present due to the plot being not random, and the points not being symmetric around 0.
A6	The error term, ϵ , is a normally distributed random variable	Q-Q plot	The normality assumption is satisfied due to the Q-Q plot displaying an approximate line at a 45° angle to the x-axis.

From **Table 12** it can be inferred that two of the regression assumptions were satisfied, however, the **issue of heteroscedasticity** is present in the third assumption. This needs to be dealt with as heteroscedasticity leads to incorrect standard errors, which prevent us from being able to quantify the uncertainty around our coefficient estimates and impact inferences. Furthermore, T-tests and F-tests become invalid, hindering the assessment of the statistical significance of the predictors as well as the explanatory power of the model. To resolve the issue, the use of robust S.E. is required.

Upon the application of the robust S.E. to our **Final Model**, we confirmed that our five predictors are highly significant (with all p values less than 1.03×10^{-9}).

Furthermore, to assess the explanatory power of the **Final Model**, a robust F-statistic was required. To obtain this, a Wald test³ was performed, and the robust F statistic was found to be 2508.6 with a p-value of less than 2.2×10^{-16} . This allowed us to conclude that, overall, the **Final Model** is highly significant and is useful for drawing inferences.

As for **multicollinearity**, all VIFs, measuring how much of the variation on the independent variable is explained by other independent variables, were found to be less than 10, indicating no multicollinearity.

5. FINAL MODEL INTERPRETATION

Upon performing the above steps and completing diagnostics, we reached the following **Final Model with statistically significant coefficient estimates based on robust S.E.:**

$$\log(\text{CHARGE}) = 5.54 + 0.0460\text{AGE} + 0.348\log(\text{BMI}) + 3.10\text{SMOKER} + 0.0889\text{REGIONS} + 0.236\text{KIDS} - 0.0379\text{AGE} * \text{SMOKER}$$

The **Final Model** coefficients suggest that all explanatory terms except for **AGE*SMOKER** positively predict **log(CHARGE)** and, therefore, **CHARGE**.

Having considered the estimate values of the **Final Model** coefficients, we argue that most of them seem plausible according to the theory and real-life references. Let us prove this point.

AGE, SMOKER, AGE*SMOKER:

Given that there is an interaction between **AGE** and **SMOKER**, the two variables as well as the interaction will be analyzed together. The coefficients of **AGE** and **SMOKER** are:

³ Mohr, F. 2018, Heteroskedasticity Robust Standard Errors in R, viewed 19 December 2021, <https://www.r-econometrics.com/methods/hcrobusterrors/>

VARIABLE	COEFFICIENT
AGE	0.0460 - 0.0379SMOKER
SMOKER	3.10 - 0.0460AGE

The coefficient of **AGE** in case an individual is a smoker (**SMOKER** = 1) is equal to 0.0081 whereas it is 0.0460 when the individual does not smoke (**SMOKER** = 0). Given that the coefficient of **AGE** is positive, in both cases, we can infer that the older a person is, the higher their medical bill is. Indeed, an individual who is one year older is closer to their life expectancy and, hence, is more expensive to insure.⁴ However, what is standing out is when an individual is a smoker the coefficient of **AGE** (0.0081) is lower compared to that of a non-smoker (0.0460). This suggests that when a smoker becomes older, they have a lower association with the medical health insurance bills as opposed to non-smokers. This point requires further investigation as it is generally perceived that a person who is older and a smoker tends to have more health issues and, thus, higher medical bills.

Similar logic can be applied while analyzing the coefficient of **SMOKER**. The coefficient of **SMOKER** is positive when the value of **AGE** is less than 67. In the Kaggle dataset which we used for the study, the ages range from 18 to 64, therefore, we can infer that the coefficient of **SMOKER** will always be positive in our case. That is, keeping all other variables fixed, an individual who is a smoker has, on average, higher medical bills because of the adverse effects smoking has both on an individual's health as well as finances⁵. However, similarly to the above, we observed that if an individual is a smoker, an increase in age has a lesser association with the medical health insurance bills.

Overall, if we had additional time and resources, we would investigate the theory and statistics behind this interaction further.

REGIONS & KIDS - Binary Variables:

Given absolute values of the coefficients for the two variables (**KIDS** - 0.236, **REGIONS** – 0.0889) we infer that having children or dependents covered by the health insurance has a stronger association with the medical costs billed by health insurance as opposed to a contractor's area of living.

An individual who has children/dependents has, on average, higher medical costs by a factor of $e^{0.236} = 1.27$. A person based in the North of the US has, on average, higher medical costs by a factor of $e^{0.0889} = 1.09$ as opposed to someone located in the South.

Log(BMI)

Keeping all other variables fixed, an increase in **BMI** is associated with a higher medical cost. Put simply, a 10% increase in **BMI** would, on average, increase individual medical costs billed by health insurance by a factor of $1.1^{0.348} = 1.034$ (equivalently, a rise in medical costs by 3.4%).

⁴ Estevez, E. 2021, How Age Affects Life Insurance Rates, viewed 19 December 2021, <https://www.investopedia.com/articles/personal-finance/022615/how-age-affects-life-insurance-rates.asp#citation-4>

⁵ Kumar, R. 2021, *Smoking Is Making You Pay 50% More Premium On A Term Insurance Plan! Here's How*, viewed 19 December 2021, <https://www.financialexpress.com/money/insurance/smoking-is-making-you-pay-50-more-premium-on-a-term-insurance-plan-heres-how/2227304/>

This is no surprise as according to one of the largest Asian insurance companies “A higher **BMI** means that the individual is at a higher risk of heart diseases and might frequently visit the hospitals due to weight-related issues⁶.” More frequent hospital visits could then translate to higher health insurance premiums.

To sum up, higher insurance premiums are associated with contractors who are older, have higher BMI, who smoke, have children/dependents and live in the Northern part of the US, please see **Table 7**.

Table 7. Factors Contributing to Higher Insurance Premiums

Factors associated with high medical costs billed by health insurance				
Older age	High BMI	Smoker	Based in North US	Has children/dependants

6. LIMITATIONS & RECOMMENDATIONS

According to the above analysis, it can be concluded that the **Final Model** not only explains the relationship between the dependent variable **CHARGE** and the five predictors along with the interaction term, but also does not violate any regression assumptions. However, while carrying out the study, some limitations were discovered.

First, as the order of data collection was not known, the assumption of the independence of the error term, ϵ , could not be tested. A suggestion for future study would be to, if possible, obtain the information on the data collection timeline and construct an index plot⁷.

Secondly, this report has only focused on the interpretation of the unstandardized coefficients, which allowed us to see the quantitative impact of a unit change in the respective independent variables on the outcome variable **CHARGE**. However, unstandardized coefficients are not as useful in revealing the relative strength of the effect of each independent variable on the response variable. That is, which of the predictors had the highest association with **CHARGE**? Thus, we suggest using the standardized coefficients⁸ and interpreting them further.

⁶ Tata AIA Life Insurance, 2021, *What is Body Mass Index and its Effect on the Life Insurance Premium*, viewed 19 December 2021, <https://www.tataaia.com/blogs/life-insurance/what-is-body-mass-index-and-its-effect-on-the-life-insurance-premium.html>

⁷ Hughes, M., Fisher, T. 2020, *Introduction to Statistical Modeling*, viewed 31 December 2021, *Chapter 8 Model Building Considerations - Miami University*<http://www.oldusers.miamioh.edu> > ...

⁸ Goyal, C. 2021, *Standardized Vs Unstandardized Regression Coefficient*, viewed 31 December 2021, <https://www.analyticsvidhya.com/blog/2021/03/standardized-vs-unstandardized-regression-coefficient/>

Finally, the analysis we carried out was only based on six predictors provided by the Kaggle dataset. These predictors may not fully explain the insurance premium that one is billed. In real life, a practitioner may want to consider a more thorough model, adding variables like family health history. Moreover, the practitioner could also consider factors including but not limited to medical history, occupation, income level and education.

CONCLUSION

TO CONCLUDE, implementation of multiple linear regression analysis, despite limitations, has successfully produced a robust and useful statistical model which sheds some light on the factors associated with the insurance premiums. Five independent terms **AGE**, **log(BMI)**, **SMOKER**, **REGIONS**, **KIDS** and the interaction **AGE*SMOKER** were found to be useful to various extents in predicting **CHARGE**, individual medical costs in \$ billed by health insurance companies. **Being older, having a higher BMI, smoking, living in the North of the US and having children/dependents** were five main factors associated with **higher** medical costs billed by health insurance providers. The proposed **Final Model** could be useful for insurance companies to assess whether a contractor is likely to have higher medical costs in the future and charge them relevant insurance premiums.

REFERENCES

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APPENDIX

Please see the R codes on page 17


```

---
title: "ST300 - Project"
output:
  html_document: default
  pdf_document: default
---

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

## Load packages for later use:
```{r}
library('ggplot2')
library('leaps')
library('car')
library("lmtest")
library("sandwich")
library("MASS")
library("visdat")
```

## Importing raw data:
```{r}
Insurance_data <- read.csv("Insurance_data_csv.csv", header = T)
#raw data with the binary variables SEX_original, SMOKER_original
recoded into SEX, SMOKER, respectively. Please note that this
recoding can be automatically done in R, but we chose to do it
manually for illustrative and interpretational purposes. REGION and
CHILDREN were recoded into REGIONS and KIDS (see Section 4.1 of the
report)
attach(Insurance_data)
View(Insurance_data) #getting an overview of the dataset
names(Insurance_data) #checking the variable names
```

## Section 1 – EXPLORATORY DATA ANALYSIS

## Summary and histograms
```{r}
summary(Insurance_data$CHARGE) #median < mean, which indicates
extreme positive values in CHARGE
hist(Insurance_data$CHARGE,breaks=100,col='blue',xlab='CHARGE',main=
'CHARGE histogram') #The data is very positively skewed, thus we
will consider logging the response variable in Section 4.3 of the
report.
```

## Scatterplots
```{r}
#CHARGE against AGE
ggplot(Insurance_data,

```

```

 aes(x = AGE, y = CHARGE)) +
 geom_point()

#CHARGE against BMI
ggplot(Insurance_data,
 aes(x = BMI, y = CHARGE)) +
 geom_point()

#AGE and SMOKER
ggplot()+
 geom_point(aes(x=AGE, y= CHARGE, col = "SMOKER"), data =
subset(Insurance_data, SMOKER == 1))+
 geom_point(aes(x=AGE, y= CHARGE, col = "NON-SMOKER"), data=
subset(Insurance_data, SMOKER == 0))+
 scale_colour_manual("", values = c("SMOKER"="blue", "NON-
SMOKER"="green"))

#Checking for missing values in our dataset
vis_miss(Insurance_data) #no missing values
```

```

Section 2 – VARIABLES SELECTION

Forwards selection:

```

```{r}
null = lm(CHARGE ~ 1, Insurance_data)
full = lm(CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SEX + SMOKER +
as.factor(REGION), data = Insurance_data)
candidate_1 = step(null, scope = list(lower = null, upper = full),
direction = "forward")
summary(candidate_1)
```

```

Forwards Selection Model:

```

CHARGE ~ CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SMOKER +
as.factor(REGION)
Removed: SEX

```

Backwards Selection:

```

```{r}
null_2 = lm(CHARGE ~ 1, Insurance_data)
full_2 = lm(CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SEX + SMOKER
+ as.factor(REGION), data = Insurance_data)
candidate_2 = step(full_2, data = Insurance_data, direction =
"backward")
summary(candidate_2)
```

```

Backwards Selection Model:

```

CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SMOKER +
as.factor(REGION)
Removed: SEX

```

```
## Stepwise Selection:
```

```
```{r}  
null_3 = lm(CHARGE ~ 1, Insurance_data)
full_3 = lm(CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SEX + SMOKER
+ as.factor(REGION), data = Insurance_data)
candidate_3 = step(null_3, scope = list(upper = full_3), data =
Insurance_data, direction="both")
summary(candidate_3)
```
```

```
## Stepwise Selection Model:
```

```
CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SMOKER +  
as.factor(REGION)  
Removed: SEX
```

```
## Best subsets:
```

```
```{r}  
candidate_4 = regsubsets(CHARGE ~ AGE + BMI + as.factor(CHILDREN) +
SEX + SMOKER + as.factor(REGION), data = Insurance_data, nbest=1)
plot(candidate_4,scale="adjr2")
plot(candidate_4,scale="bic")
par(mfrow = c(1,2))
```
```

SEX was removed in all of the potential models under both criteria hence:

```
## Best subsets Model:
```

```
CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SMOKER +  
as.factor(REGION)  
Removed: SEX
```

```
## Section 3 – ANALYSIS OF FOUR CANDIDATE MODELS
```

```
## Candidate Model 1: log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS +  
SMOKER + REGIONS (done by 33847)
```

Having performed variable selection, the baseline model was found to be

CHARGE ~ AGE + BMI + as.factor(CHILDREN) + SMOKER +
as.factor(REGION). CHILDREN was then transformed into a binary
variable KIDS and the new baseline model is below:

```
## BASELINE MODEL:
```

```
CHARGE ~ AGE + BMI + KIDS + SMOKER + as.factor(REGION)  
```{r}  
Baseline_model = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER +
as.factor(REGION), data = Insurance_data)
summary(Baseline_model)$r.squared
summary(Baseline_model)$adj.r.squared
```
```

```
## BEST SUBSETS REGRESSION:
```

```

```{r}
model = regsubsets(CHARGE ~ AGE + as.factor(SEX) + BMI +
as.factor(SMOKER) + as.factor(REGION) + as.factor(KIDS), data =
Insurance_data, nbest=1)
plot(model,scale="adjr2")
plot(model,scale="bic")
par(mfrow = c(1,2))
```

```

Observing the first plot, we see that best subsets regression suggests removing REGION northwest, however this is not possible as it is a level of the categorical variable REGION.

```

```{r}
summary(Baseline_model)
```

```

Moreover Region Northwest is insignificant.

Therefore, we try regrouping the variable REGION into binary variable REGIONS which takes 1 if the individual is based in Northeast or Northwest US, and 0 if otherwise.

Therefore the new model is:

```

```{r}
Model_1 = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS, data =
Insurance_data)
summary(Model_1)
```

```

Trying Transformations:

```

```{r}
hist(CHARGE)
```

```

We see that CHARGE, the dependent variable is relatively skewed.

Using the log transformation:

```

```{r}
hist(log(CHARGE))
```

```

We see that this helps with the skew of CHARGE and therefore we use log(CHARGE) instead:

```

```{r}
Model_2 = lm(log(CHARGE) ~ AGE + BMI + KIDS + SMOKER + REGIONS, data
= Insurance_data)
summary(Model_2)
```

```

This leads to an increase in adjusted R squared and multiple R squared therefore this model is retained.

Looking at correlations between log(CHARGE) and transformed

dependent variables:

Correlation between CHARGE and AGE (with different transformations):

```
```{r}
cor(log(CHARGE), AGE)
cor(log(CHARGE),exp(AGE))
cor(log(CHARGE),sqrt(AGE)) #Second Highest correlation
cor(log(CHARGE),log(AGE)) #Highest correlation
cor(log(CHARGE),AGE^2)
```
```

Trying sqrt transformation:

```
```{r}
Model_3 = lm(log(CHARGE) ~ sqrt(AGE) + BMI + KIDS + SMOKER +
REGIONS, data = Insurance_data)
summary(Model_3)
```
```

This leads to an increase in adjusted R squared and multiple R squared therefore this is potentially a good transformation!

Correlation between CHARGE and BMI (with different transformations):

```
```{r}
cor(log(CHARGE), BMI) #Second Highest correlation
cor(log(CHARGE),exp(BMI))
cor(log(CHARGE),sqrt(BMI)) #Second Highest correlation
cor(log(CHARGE),log(BMI)) #Highest correlation
cor(log(CHARGE),BMI^2)
```
```

Trying log transformation:

```
```{r}
Model_4 = lm(log(CHARGE) ~ AGE + log(BMI) + KIDS + SMOKER + REGIONS,
data = Insurance_data)
summary(Model_4)
```
```

This leads to an increase in adjusted R squared and multiple R squared therefore this is potentially a good transformation!

Combining good transformations

All transformations are compared to Model_2:

Model_2: $\log(\text{CHARGE}) \sim \text{AGE} + \text{BMI} + \text{KIDS} + \text{SMOKER} + \text{REGIONS}$. Multiple R-squared: 0.7628, Adjusted R-squared: 0.7619

Model_3: Model_2 but with $\sqrt{\text{AGE}}$. Multiple R-squared: 0.7643, Adjusted R-squared: 0.7635

Model_4: Model_2 but with $\log(\text{BMI})$. Multiple R-squared: 0.7634, Adjusted R-squared: 0.7626

Combining transformations from Model_2, Model_3 and Model_4:

```

```{r}
Model_5 = lm(log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS + SMOKER +
REGIONS, data = Insurance_data)
summary(Model_5)
```

```

This yields a higher Multiple R-squared and Adjusted R-squared than all other models, suggesting that this is a good model.

Trying interaction terms on Model_5:

```

log(BMI)*SMOKER:
```{r}
Model_6 = lm(log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS + SMOKER +
REGIONS + log(BMI)*SMOKER, data = Insurance_data)
summary(Model_6)$r.squared
summary(Model_6)$adj.r.squared
```

```

This leads to a large increase in Multiple R-squared and Adjusted R-squared

Lets check VIF:

```

```{r}
vif(Model_6)
```

```

There seems to be a large issue of Multicollinearity.

```

log(BMI)*KIDS:
```{r}
Model_7 = lm(log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS + SMOKER +
REGIONS + log(BMI)*KIDS, data = Insurance_data)
summary(Model_7)$r.squared
summary(Model_7)$adj.r.squared
```

```

This leads to a increase in Multiple R-squared and Adjusted R-squared

Lets check VIF:

```

```{r}
vif(Model_7)
```

```

Again, there seems to be a large issue of Multicollinearity.

```

log(BMI)*sqrt(AGE):
```{r}
Model_8 = lm(log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS + SMOKER +
REGIONS + log(BMI)*sqrt(AGE), data = Insurance_data)
summary(Model_8)$r.squared
summary(Model_8)$adj.r.squared
```

```

This does not lead to an increase in Adjusted R-squared

CONCLUSION:

Although adding interactions makes theoretical sense, they do not seem to improve the model. Either, they introduces multicollinearity, or do not help improve model characteristics such as Adjusted R-squared.

Hence we can conclude that the best model from the candidate models tried above (Model_1 to Model_8) is Model_5:

```
## Model_5: log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS + SMOKER +  
REGIONS (a.k.a. Candidate Model 1)
```

Please note that many steps undertaken when searching for Candidate Models 2 and 3 were similar to those described for Candidate Model 1, so a lot of the codes and comments were omitted to avoid repetition.

```
## Candidate Model 2: log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS +  
SMOKER + as.factor(REGIONS) (done by 31380)
```

```
```{r}  
Baseline_model = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER +
as.factor(REGION), data = Insurance_data)
summary(Baseline_model) #obtain general information about the
baseline model which has the binary variable KIDS rather than the
categorical variable CHILDREN
```
```

As indicated in the EDA, CHARGE is found to be skewed, hence, transformations should be applied to it.

```
```{r}  
hist(CHARGE) #baseline
hist(log(CHARGE))
hist(sqrt(CHARGE))
boxcox(lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + as.factor(REGION),
data = Insurance_data), data = Insurance_data) #this was not covered
in the course and it would be hard to justify the transformation
suggested by the Box-Cox method, so choose not to use it.
```
```

Upon the visual inspection of the above plots, the logarithmic transformation is chose as it results in a distribution which is closest to looking Normal.

Based on an argument similar to that provided for Candidate Model 2, sqrt(AGE) and log(BMI) were selected instead of AGE and BMI, in turn. The resulting model is:

```
```{r}  
Candidate_Model_2 = lm(log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS +
SMOKER + as.factor(REGION), data = Insurance_data)
```

```
summary(Candidate_Model_2)
```
```

Note that even though REGION northwest is insignificant, it cannot be removed as it represents a level of the variable REGION. Moreover, the other two dummy variables associated with REGION are significant, so we choose to retain REGION as a variable.

```
## Trying interaction terms on Candidate_Model_2:
```

```
```{r}
pairs(~log(CHARGE) + sqrt(AGE) + log(BMI) + KIDS + SMOKER, data =
Insurance_data)
```
```

A visual inspection suggests that there are no obvious patterns, and, hence, interactions between the predictors.

CONCLUSION:

Having introduced transformations to both the independent and dependent variables, the below candidate model was reached. Please note that had this model been selected as the final one, further analysis and checks would need to be undertaken to ensure that the regression assumptions were met and the model made practical sense.

```
## log(CHARGE) ~ sqrt(AGE) + log(BMI) + KIDS + SMOKER +
as.factor(REGIONS) (Candidate Model 2)
```

```
## Candidate Model 3: log(CHARGE) ~ sqrt(AGE) + log(BMI) + CHILDREN
+ SMOKER + REGION (done by 35677)
```

```
```{r}
C1 = lm(log(CHARGE) ~ log(AGE) + SEX + BMI + SMOKER +
as.factor(REGION) + as.factor(CHILDREN), data = Insurance_data)
summary(C1) #baseline model
```
```

```
##C2: log(charge), sqrt(age), sex, smoker, region/children fac
```

```
```{r}
C2 = lm(log(CHARGE) ~ sqrt(AGE) + SEX + BMI + SMOKER +
as.factor(REGION) + as.factor(CHILDREN), data = Insurance_data)
summary(C2)
```
```

```
##C3: log(charge), sqrt(age), sqrt(BMI), smoker, region/children fac
```

```
```{r}
C3 = lm(log(CHARGE) ~ sqrt(AGE) + sqrt(BMI) + SMOKER +
as.factor(REGION) + as.factor(CHILDREN), data = Insurance_data)
summary(C3)
```
```

Notice that using as.factor(CHILDREN) increases the number of dummy variables.

```
##C4: log(charge), sqrt(age), smoker, region fac, treating Children
```


as a continuous variable

```
```{r}
C4 = lm(log(CHARGE) ~ sqrt(AGE) + BMI + SMOKER + as.factor(REGION) +
CHILDREN, data = Insurance_data)
summary(C4)
```
```

##C5: log (charge), sqrt(age), log(BMI), sex, smoker, region fac,
treating Children as a continuous variable

```
```{r}
C5 = lm(log(CHARGE) ~ sqrt(AGE) + SEX + log(BMI) + SMOKER +
as.factor(REGION) + CHILDREN, data = Insurance_data)
summary(C5)
```
```

Notice that the variable northwest is always insignificant, we
regroup the region into a dummy variable

##C6: log(charge), BMI, sqrt(age), smoker, region, treating Children
as a continuous variable

```
```{r}
C6 = lm(log(CHARGE) ~ sqrt(AGE) + log(BMI) + SMOKER + CHILDREN +
REGIONS, data = Insurance_data)
summary(C6)
```
```

CONCLUSION:

C6 has the best values of multiple R^2 and adjusted R^2 . However, it
treats CHILDREN as a continuous variable, which may not make
practical sense.

log(CHARGE) ~ sqrt(AGE) + log(BMI) + SMOKER + CHILDREN + REGIONS
(a.k.a Candidate Model 3)

Candidate Model 4: log(CHARGE) ~ AGE + log(BMI) + KIDS + SMOKER +
REGION + AGE*SMOKER (done by 34329)

This was chosen to be our final model. The code for the model as
well as the detailed explanation of what steps were undertaken to
reach this model and further improve it, can be found under Section
4.

Section 4 – FINAL MODEL DERIVATION & DIAGNOSTICS

FINAL MODEL:

log(CHARGE) ~ AGE + log(BMI) + SMOKER + REGIONS + KIDS + AGE*SMOKER

Section 4.2 – INTERACTIONS

```
## BASELINE MODEL:
CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS
```{r}
Baseline_model = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS,
data = Insurance_data)
summary(Baseline_model)$r.squared
summary(Baseline_model)$adj.r.squared
vif(Baseline_model)
```
```

```
## Interaction 1: AGE*BMI
```{r}
Interaction_1 = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS +
AGE*BMI, data = Insurance_data)
summary(Interaction_1)$r.squared
summary(Interaction_1)$adj.r.squared
vif(Interaction_1)
```
```

This leads to multicollinearity as well as a decline in adjusted R squared.
Hence this is not a good interaction.

```
## Interaction 2: AGE*SMOKER
```{r}
Interaction_2 = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS +
AGE*SMOKER, data = Insurance_data)
summary(Interaction_2)$r.squared
summary(Interaction_2)$adj.r.squared
vif(Interaction_2)
```
```

This leads to an increase in R squared and adjusted R squared
VIF increases for SMOKER and AGE*SMOKER, however it is still under the threshold (10).
Hence this is potentially a good interaction.

```
## Interaction 3: BMI*KIDS
```{r}
Interaction_3 = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS +
BMI*KIDS, data = Insurance_data)
summary(Interaction_3)$r.squared
summary(Interaction_3)$adj.r.squared
vif(Interaction_3)
```
```

This leads to an increase in R squared and adjusted R squared.
VIF is above 10 for KIDS and BMI*KIDS, introduces multicollinearity.
Hence this is not a good interaction.

```
## Interaction 4: BMI*SMOKER
```{r}
Interaction_4 = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS +
BMI*SMOKER, data = Insurance_data)
```

```
summary(Interaction_4)$r.squared
summary(Interaction_4)$adj.r.squared
vif(Interaction_4)
```

```

This leads to a significant increase in R squared and adjusted R squared

However, VIF is above 10 for SMOKER and BMI:SMOKER, introduces multicollinearity

Hence this is not a good interaction.

Now let's apply the AGE*SMOKER interaction to the Baseline Model.

```
```{r}
CURRENT_MODEL = lm(CHARGE ~ AGE + BMI + KIDS + SMOKER + REGIONS +
AGE*SMOKER)
summary(CURRENT_MODEL)$r.squared
summary(CURRENT_MODEL)$adj.r.squared
vif(CURRENT_MODEL)
```

```

Section 4.3 – TRANSFORMATIONS

Transforming the response variable CHARGE

```
```{r}
hist(CHARGE) #the response variable CHARGE is heavily skewed
(positive)
hist(log(CHARGE)) #apply log transformation to CHARGE to get a more
bell-shaped symmetric distribution
```

```

Transforming the predictor variable BMI

```
```{r}
cor(log(CHARGE), 1/(BMI)) #~0.140 – largest absolute value of the
correlation achieved. However, it is complicated to interpret
practically and justify the reciprocal transformation of a variable,
so we do not choose this transformation.
cor(log(CHARGE), log(BMI)) #~0.137 – second largest absolute value
of the correlation achieved – choose this transformation.
cor(log(CHARGE), sqrt(BMI))
cor(log(CHARGE), (BMI))
cor(log(CHARGE), (BMI)^2)
cor(log(CHARGE), exp(BMI)) #least useful transformation based on the
absolute value of the correlation
```

```

Now let's apply the transformations to the CURRENT MODEL above (with the interaction AGE*SMOKER):

```
```{r}
CURRENT_MODEL = lm(log(CHARGE) ~ AGE + log(BMI) + KIDS + SMOKER +
REGIONS + AGE*SMOKER)
summary(CURRENT_MODEL)$r.squared
```

```

```
summary(CURRENT_MODEL)$adj.r.squared
```
```

```
Section 4.4 – OUTLIERS, LEVERAGE POINTS & INFLUENTIAL POINTS
```

```
```{r}
```

```
#Filtering the outliers
```

```
plot(rstandard(CURRENT_MODEL))
```

```
options(max.print=1000000)
```

```
max(rstandard(CURRENT_MODEL))
```

```
sort(rstandard(CURRENT_MODEL))#We observe that there are indeed  
values above 3. These are filtered below
```

```
#Filtering the high leverage points
```

```
plot(hatvalues(CURRENT_MODEL))
```

```
#k chosen to be 3
```

```
 #(p/n) * k  $\approx$  0.013
```

```
hatvalues(CURRENT_MODEL)
```

```
sort(hatvalues(CURRENT_MODEL))
```

```
max(hatvalues(CURRENT_MODEL)) #We observe that there are indeed  
values above 0.013. These are filtered below
```

```
#Identifying influential points
```

```
cooks.distance(CURRENT_MODEL)
```

```
plot(cooks.distance(CURRENT_MODEL))
```

```
max(cooks.distance(CURRENT_MODEL))#We observe that there are indeed  
values above  $\approx$  0.003. These are filtered below
```

```
```
```

```
Filtering the outliers above 3:
```

```
```{r}
```

```
outliers_threshold = 3
```

```
rstandard(CURRENT_MODEL)
```

```
[rstandard(CURRENT_MODEL)>outliers_threshold]
```

```
```
```

```
Filtering the high leverage points above 0.013:
```

```
```{r}
```

```
high_leverage_threshold = 3*(6/1338)
```

```
hatvalues(CURRENT_MODEL)
```

```
[hatvalues(CURRENT_MODEL)>high_leverage_threshold]
```

```
```
```

```
Filtering the influential points:
```

```
```{r}
```

```
influential_points_threshold = 4/(1338)
```

```
cooks.distance(CURRENT_MODEL)
```

```
[cooks.distance(CURRENT_MODEL)>influential_points_threshold]
```

```
```
```

```
OUTLIERS, LEVERAGE POINTS & INFLUENTIAL POINTS PLOTS:
```

```
```{r}
```

```
#OUTLIER POINTS PLOT (With threshold as >3)
```

```
ggplot(aes(seq_along(rstandard(CURRENT_MODEL)),
```

```
rstandard(CURRENT_MODEL)), data=Insurance_data) + geom_point() +
```

```

geom_hline(yintercept = 3, linetype='dashed' , color='red')+
xlab("Observation number")+
ylab("Standardised residuals")

#HIGH LEVERAGE POINTS PLOT
high_leverage_threshold = (6/1338)*3
ggplot(aes(seq_along(hatvalues(CURRENT_MODEL)),
hatvalues(CURRENT_MODEL)), data=Insurance_data) + geom_point() +
geom_hline(yintercept = high_leverage_threshold, linetype='dashed' ,
color='red')+ xlab("Observation number")+
ylab("Hat Values")

#INFLUENTIAL POINTS PLOT
threshold = 4/1338
ggplot(aes(seq_along(cooks.distance(CURRENT_MODEL)),
cooks.distance(CURRENT_MODEL)), data=Insurance_data) +
  geom_col()+
  geom_hline(yintercept = threshold, linetype='dashed' ,
color='red') + xlab("Observation number") +
  xlab("Observation number")+
  ylab("Cook's Distance")

#INFLUENTIAL POINTS DOTS PLOT
influential_points_threshold = 4/1338
ggplot(aes(seq_along(cooks.distance(CURRENT_MODEL)),
cooks.distance(CURRENT_MODEL)), data=Insurance_data) + geom_point()
+
geom_hline(yintercept = influential_points_threshold,
linetype='dashed' , color='red')+ xlab("Observation number")+
ylab("Cook's distance")
```

```

```

Let's import the edited dataset with influential points removed:
```{r}
detach(Insurance_data)
rm(Insurance_data) #remove the dataset to avoid confusion
Insurance_data_points_removed <-
read.csv("Insurance_data_points_removed_csv.csv", header = T) #data
with influential points removed.
attach(Insurance_data_points_removed)
View(Insurance_data_points_removed) #getting an overview of the
dataset
names(Insurance_data_points_removed) #checking the variable names
```

```

```

After removing the influential points, the final model is now:
```{r}
Final_model = lm(log(CHARGE) ~ AGE + log(BMI) + SMOKER + REGIONS +
KIDS + AGE*SMOKER, data = Insurance_data_points_removed)
```

```

## Section 4.5 – ADJUSTED R<sup>2</sup>, RESIDUAL ANALYSIS & MULTICOLLINEARITY

```

```{r}
Final_model = lm(log(CHARGE) ~ AGE + log(BMI) + SMOKER + REGIONS +
KIDS + AGE*SMOKER, data = Insurance_data_points_removed)
summary(Final_model)$adj.r.squared
```

```

```

Checking Assumption 1 – Linearity
```{r}
plot(rstandard(Final_model), AGE)
plot(rstandard(Final_model), log(BMI))
```

```

```

Checking Assumption 4: Homoscedasticity and Assumption 6:
Normality
```{r}
#Check for heteroscedasticity by looking at residuals vs fitted
values & for normality by looking at the q-q plot:
plot(rstandard(Final_model), fitted(Final_model))
plot(Final_model)
```

```

Heteroscedasticity seems to be an issue. Therefore Robust SE will be used.

```

T-Tests under robust S.E.:
```{r}
coefTest(Final_model, vcov = vcovHC(Final_model, type = "HC0"))
```

```

Under Robust SE, the independent variables are found to be highly significant

```

F-Tests:
```{r}
null_model = lm(log(CHARGE)~1,Insurance_data_points_removed)
Final_model = lm(log(CHARGE) ~ AGE + log(BMI) + SMOKER + REGIONS +
KIDS + AGE*SMOKER, data = Insurance_data_points_removed)
```

```

```

A Wald test was performed to obtain a robust F-statistic
```{r}
waldtest(null_model, Final_model, vcov = vcovHC(Final_model, type =
"HC0"))
```

```

The model was found to be highly significant – the robust F statistic has a p-value of less than  $2.20 \times 10^{-16}$ .

#### ## Section 4.7 – MULTICOLLINEARITY

```

```{r}
vif(Final_model)
```

```

All VIFs were less than 10, indicating no multicollinearity.

## ## Section 5 – FINAL MODEL INTERPRETATION

THE FINAL MODEL:

\$\$

$\log(\text{CHARGE}) = 5.54 + 0.0460\text{AGE} + 0.348\log(\text{BMI}) + 3.10\text{SMOKER} + 0.0889\text{REGIONS} + 0.236\text{KIDS} - 0.0379\text{AGE}*\text{SMOKER}$

\$\$

SUMMARY STATISTICS OF THE FINAL MODEL:

```{r}

```
Final_model = lm(log(CHARGE) ~ AGE + log(BMI) + SMOKER + REGIONS + KIDS + AGE*SMOKER, data = Insurance_data_points_removed)
summary(Final_model)$adj.r.squared
coeftest(Final_model, vcov = vcovHC(Final_model, type = "HC0"))
null_model = lm(log(CHARGE)~1,Insurance_data_points_removed)
waldtest(null_model, Final_model, vcov = vcovHC(Final_model, type = "HC0"))
```
```

Adjusted R squared was found to be 0.923.

Under Robust SE, the independent variables were found to be highly significant

The robust F statistic was found to be 2508.6 with a p-value of less than  $2.2 \times 10^{-16}$ . Hence the overall Final Model is highly significant.

Please note:

Candidate Model 1 was developed by 33847

Candidate Model 2 was developed by 31380

Candidate Model 3 was developed by 35677

Candidate Model 4 was developed by 34329, yet contributions from the other 3 team members were made to finalize the model.

Overall, all team members collaborated to ensure equal contribution to this markdown's production :)